Sample size against variance heterogeneity in the denominator degrees of freedom of Welch's ANOVA

Draft for the appendix of visStatistics — not yet included in the vignette

This appendix separates the contribution of the group sizes from that of the variances in the denominator degrees of freedom of Welch's ANOVA, in balanced and in unbalanced designs.

Throughout, k is the number of groups, $i = 1, \ldots, k$ the group index, n_i the size of group i and σ_i^2 its population variance. The denominator degrees of freedom of Eq. (**??**), as returned by `oneway.test()`, are

$$\nu = \frac{k^2 - 1}{3 \displaystyle\sum_{i=1}^{k} \frac{\left(1 - w_i/w\right)^2}{n_i - 1}}, \qquad w_i = \frac{n_i}{s_i^2}, \quad w = \sum_{i=1}^{k} w_i. \tag{1}$$

The quantity examined here is Eq. (1) with each s_i^2 replaced by σ_i^2, which is fixed by the design (n_i, σ_i) alone and therefore comparable across configurations, whereas the computed value varies from replication to replication. No claim is made below about how closely the computed value tracks it at small n_i.

Separating the group sizes from the variances

Let $\bar{n} = k^{-1} \sum_{i=1}^{k} n_i$ denote the mean group size and write

$$n_i = c_i\, \bar{n}, \qquad \sum_{i=1}^{k} c_i = k, \tag{2}$$

with design constants c_i fixing the shape of the size vector while $\bar{n}$ carries its scale. In the simulations of Section **??**, $\mathbf{c} = (0.5,\, 0.8,\, 1.2,\, 1.5)$ for the unbalanced designs and $\mathbf{c} = (1, 1, 1, 1)$ for the balanced ones.

Substituting Eq. (2) into the weights of Eq. (1), the factor $\bar{n}$ appears in w_i and in w alike and cancels in their ratio, so that the normalised weights

$$u_i = \frac{w_i}{w} = \frac{c_i/\sigma_i^2}{\displaystyle\sum_{j=1}^{k} c_j/\sigma_j^2}, \qquad \sum_{i=1}^{k} u_i = 1, \tag{3}$$

depend on the variances and on the shape of the size vector, but not on $\bar{n}$. With Eq. (3),

$$\nu = \frac{k^2 - 1}{3 \displaystyle\sum_{i=1}^{k} \frac{(1 - u_i)^2}{c_i \bar{n} - 1}}, \tag{4}$$

exactly. The mean group size now enters only through the denominators $c_i \bar{n} - 1$.

Balanced designs

For $c_i = 1$, $i = 1, \dots, k$, every denominator term of Eq. (4) carries the same factor $\bar{n} - 1$, which can therefore be taken out of the sum. Writing

$$D = \frac{k^2 - 1}{3 \sum_{i=1}^{k} \frac{(1 - u_i)^2}{c_i}}, \tag{5}$$

which does not depend on $\bar{n}$, Eq. (4) becomes

$$\nu = D\,(\bar{n} - 1). \tag{6}$$

Unbalanced designs

For general c_i write $c_i\bar{n} - 1 = c_i\bar{n}\,(1 - (c_i\bar{n})^{-1})$ and expand the reciprocal as a geometric series in $(c_i\bar{n})^{-1}$, which converges provided

$$c_i\,\bar{n} > 1 \qquad \text{for all } i = 1, \dots, k. \tag{7}$$

In the simulated designs $\min_i c_i = 0.5$, so Eq. (7) holds from $\bar{n} = 3$ on. The sum in Eq. (4) becomes

$$\sum_{i=1}^{k} \frac{(1 - u_i)^2}{c_i\bar{n} - 1} = \frac{1}{\bar{n}} \sum_{i=1}^{k} \frac{(1 - u_i)^2}{c_i} \left(1 + \frac{1}{c_i\bar{n}} + O(\bar{n}^{-2})\right) = \frac{1}{\bar{n}} \left(\frac{k^2 - 1}{3D} + \frac{E}{\bar{n}} + O(\bar{n}^{-2})\right), \tag{8}$$

with D from Eq. (5) and

$$E = \sum_{i=1}^{k} \frac{(1 - u_i)^2}{c_i^2}. \tag{9}$$

Inserting Eq. (8) into Eq. (4) and expanding the reciprocal once more gives

$$\nu = D\,\bar{n} \;-\; \frac{3D^2E}{k^2 - 1} \;+\; O(\bar{n}^{-1}). \tag{10}$$

Eq. (6) is the special case $c_i \equiv 1$, for which $E = (k^2 - 1)/(3D)$ and the constant term reduces to D.

The ordering of the two influences

The variances enter Eq. (10) only through D, which contains no $\bar{n}$. Since $\sum_i u_i = 1$ gives $\sum_i (1 - u_i) = k - 1$, the Cauchy–Schwarz inequality in the form $\left(\sum_i a_i\right)^2 \le \left(\sum_i a_i^2/c_i\right)\left(\sum_i c_i\right)$ applied to $a_i = 1 - u_i$ yields

$$\sum_{i=1}^{k} \frac{(1 - u_i)^2}{c_i} \;\ge\; \frac{(k - 1)^2}{k}, \tag{11}$$

with equality if and only if $(1 - u_i)/c_i$ is constant in i, which holds in particular for a balanced design with equal variances. In the opposite extreme one weight tends to one and the others to zero, so that $\sum_i (1 - u_i)^2/c_i \to \sum_{i \ne j} 1/c_i$. Hence

$$\frac{k^2-1}{3\sum_{i=1}^{k} 1/c_i} \;\le\; D \;\le\; \frac{k(k+1)}{3(k-1)}. \tag{12}$$

The two contributions are thus of different kind. The group sizes enter as an unbounded scale,

$$\lim_{\bar{n}\to\infty} \frac{\nu}{\bar{n}} = D, \tag{13}$$

by Eq. (10), whereas the variances enter only through D, confined by Eq. (12) to a fixed interval whose upper end is attained under homoscedasticity in a balanced design. No configuration of $\sigma_1, \dots, \sigma_k$ prevents ν from growing proportionally to $\bar{n}$.